

Vector

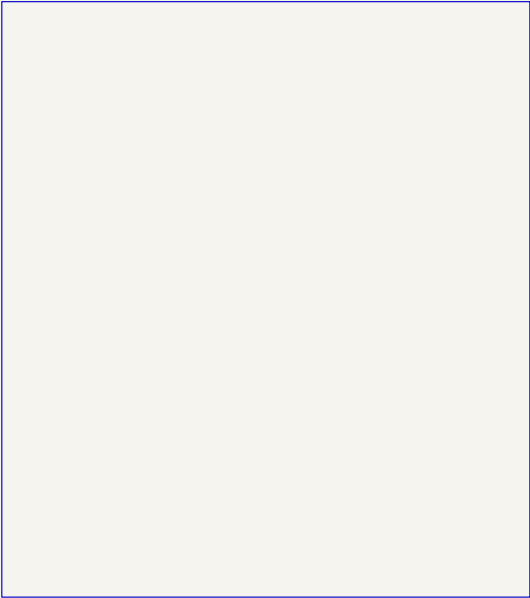
Variant: DRAFT

2026-07-06

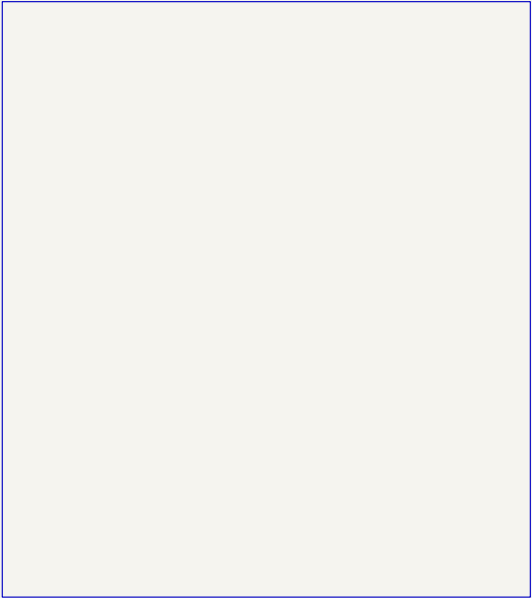
Rev 1.0.0

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TOP VIEW



BOTTOM VIEW



NOTES

Comment

Not fitted components are marked as **X**

DRAFT - Very early stage of schematic, ignore details.

PRELIMINARY - Close to final schematic.

CHECKED - There shouldn't be any mistakes. Contact the engineer if you find any.

RELEASED - A board with this schematic has been sent to production.

DESIGN CONSIDERATIONS

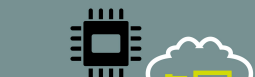
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Example text for
informational design
notes.

DESIGN NOTE:
Example text for
debug notes.

DESIGN NOTE:
Example text for
cautionary design
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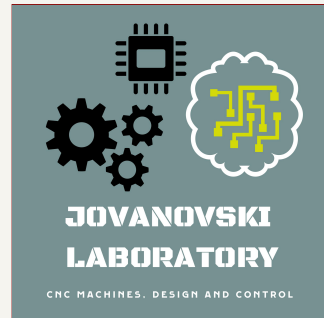
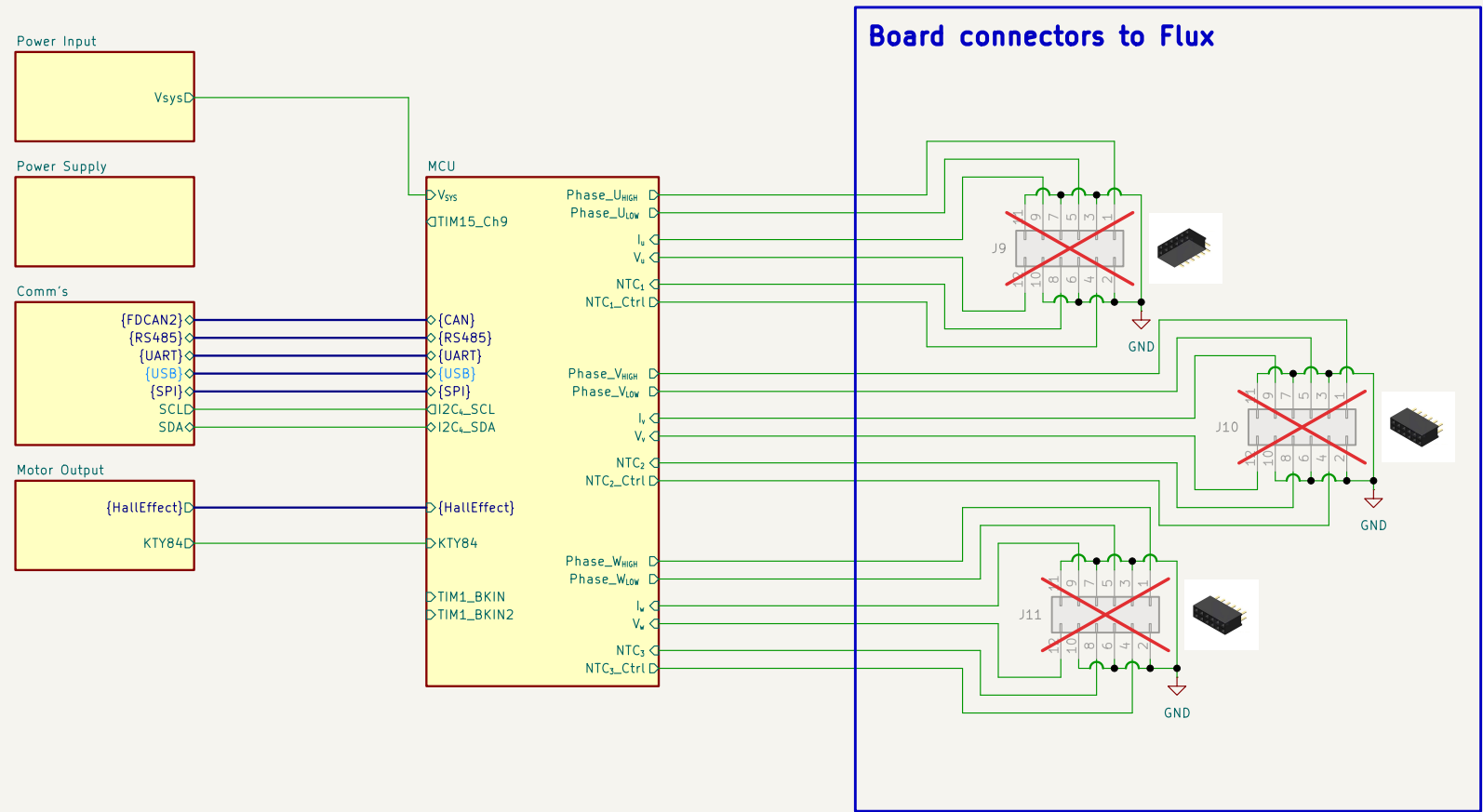
DESIGN NOTE:
Example text for
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LAYOUT NOTE:
Example text for
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 JOVANOVSKI LABORATORY CNC MACHINES. DESIGN AND CONTROL	Comments:		Company: Jovanovski Laboratory - JLab		Variant:	
			Board Name: Vector		Project Name: Vortex	
	Sheet Title:	File Name: PCBA-VECTOR-01A.kicad_sch	Designer: Bvmy	Date:		Revision: 1.0.0
	Sheet Path: /		Reviewer:		Size: A3	Sheet: 1 of 9

1		2		3		4		5			

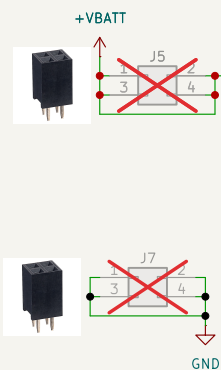
[3] Project Architecture



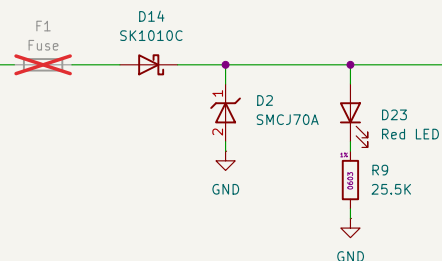
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Sheet Title:	File Name: ProjectArchitecture.kicad_sch	Designer: Bvmy	Date:	Revision: 1.0.0
	Sheet Path: /Project Architecture/		Size: A4	Sheet: 3 of 9

[4] Power Input

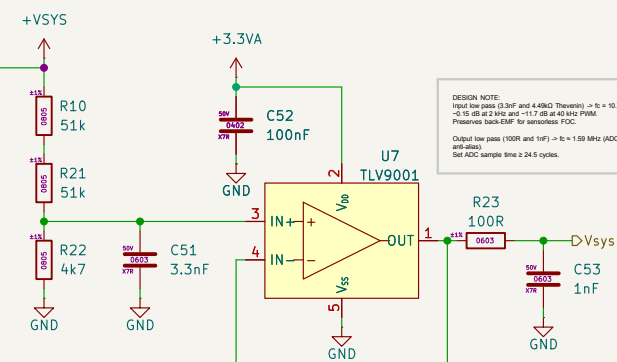
Board connectors for power to Flux



Input protection



Battery Voltage



Comments:

Company:

Jovanovski Laboratory - JLab

Variant:

Board Name:

Vector

Project Name:

Vortex

Sheet Title:

File Name:

PowerInput.kicad_sch

Designer:

Bvmy

Date:

Revision:

1.0.0

Sheet Path:

/Project Architecture/Power Input/

Reviewer:

Size:

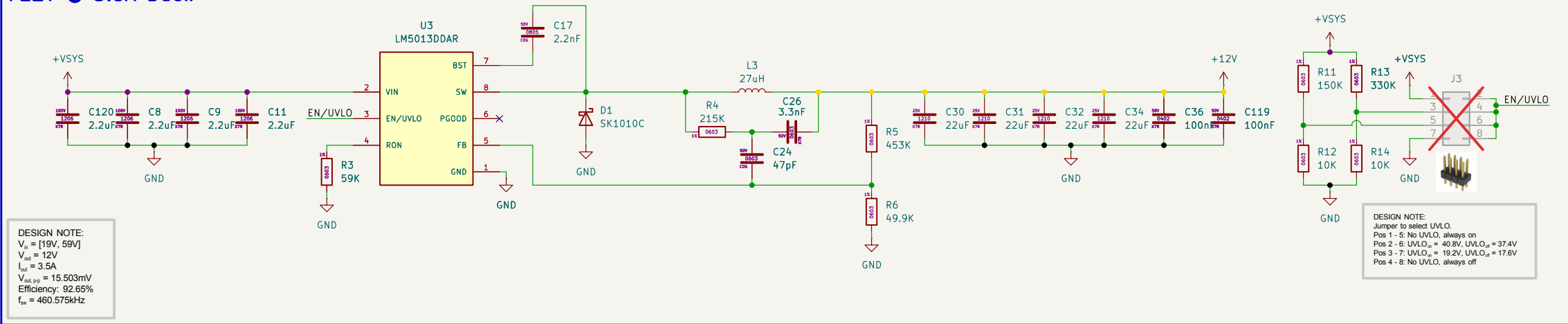
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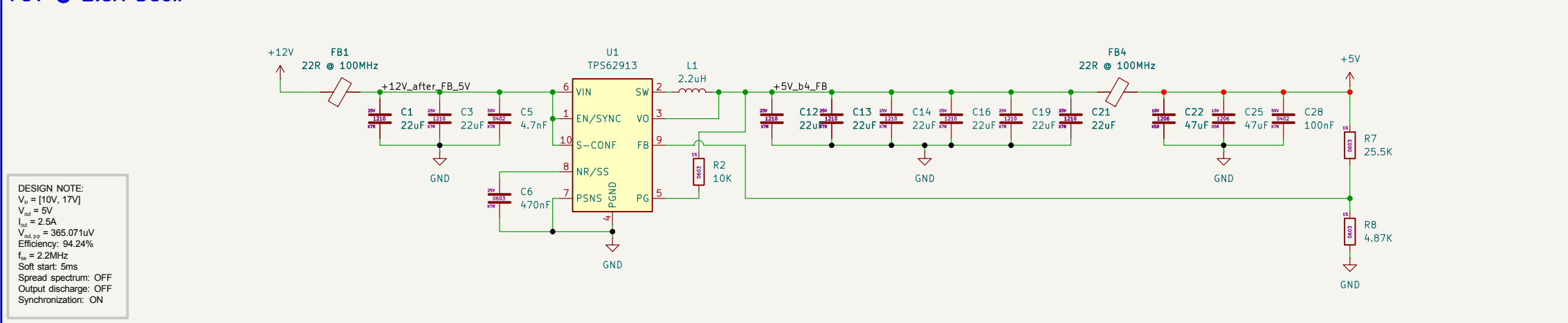
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[5] Power Supply

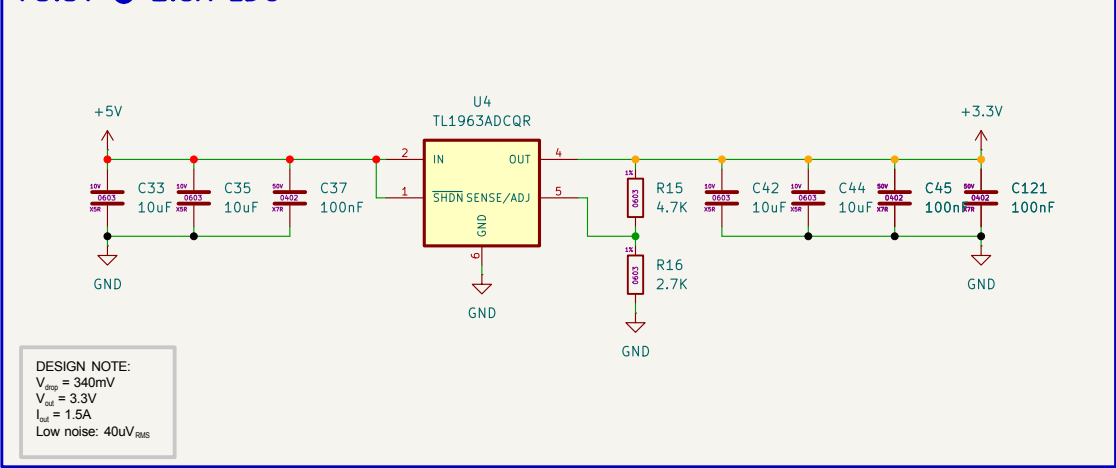
+12V • 3.5A Buck



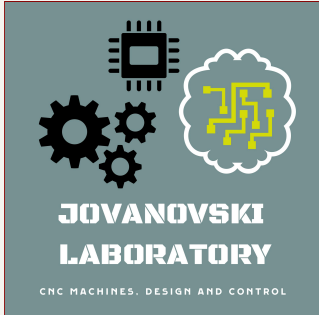
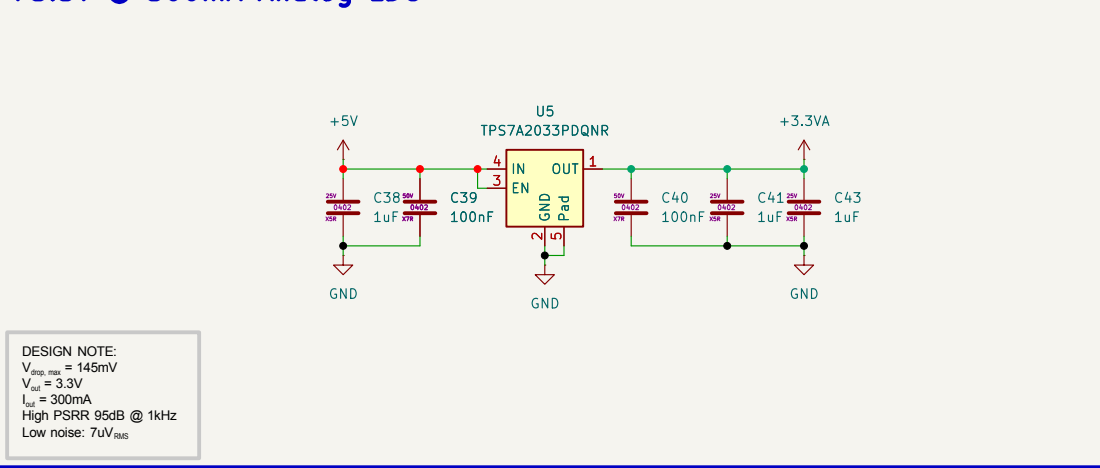
+5V • 2.5A Buck



+3.3V • 1.5A LDO

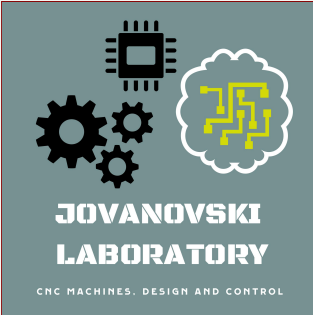
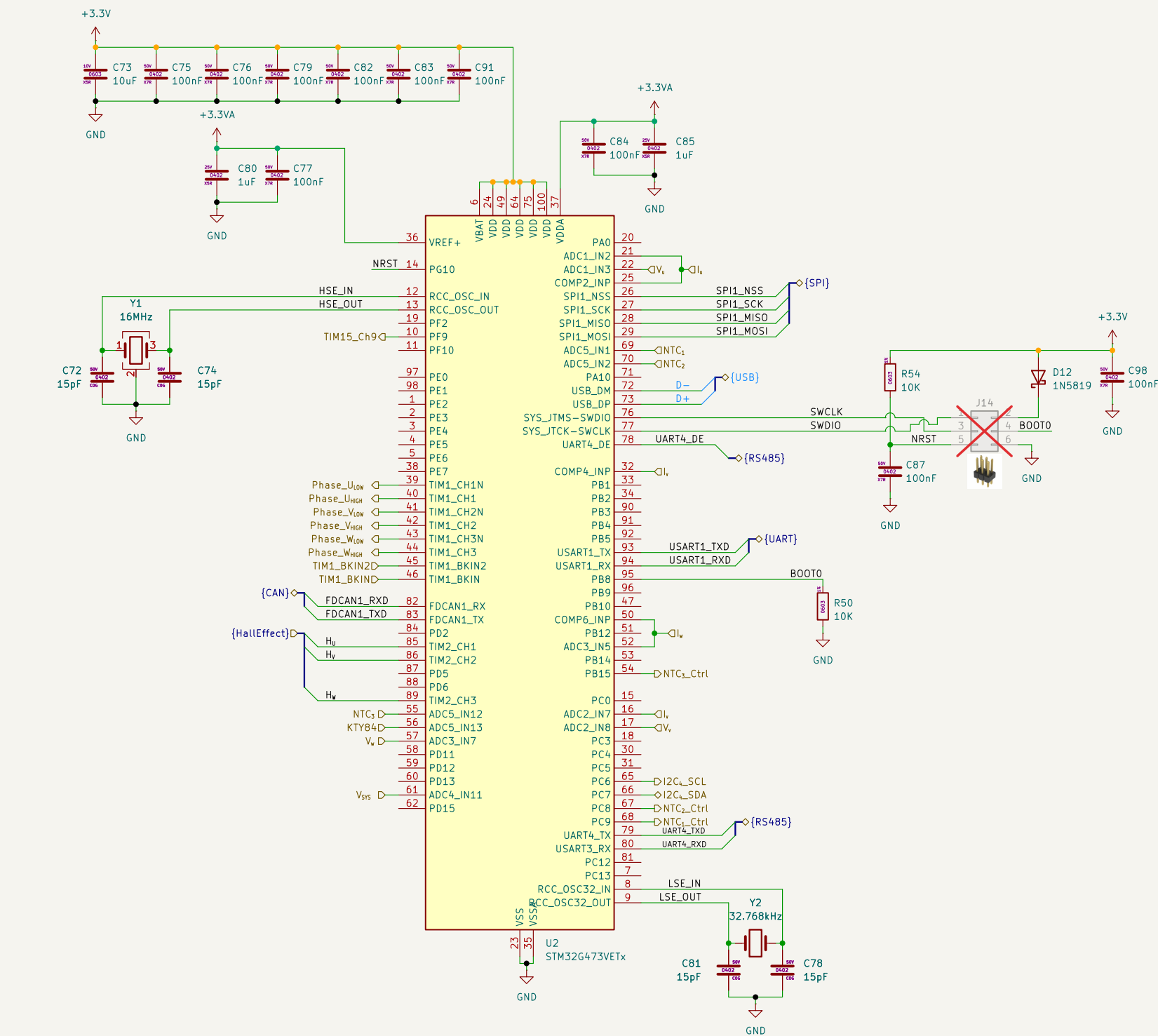


+3.3V • 300mA Analog LDO



Comments:		Company: Jovanovski Laboratory - JLab		Variant:	
Sheet Title:		Board Name: Vector		Project Name: Vortex	
Sheet Path: /Project Architecture/Power Supply/		File Name: PowerSupply.kicad_sch	Designer: Bvmy	Date:	Revision: 1.0.0
		Reviewer:	Size: A3	Sheet: 5 of 9	

[6] MCU



Comments:		Company: Jovanovski Laboratory - JLab		Variant:	
		Board Name: Vector		Project Name: Vortex	
Sheet Title:		File Name: MCU.kicad_sch	Designer: Bvmy	Date:	Revision: 1.0.0
Sheet Path: /Project Architecture/MCU/			Reviewer:	Size: A3	Sheet: 6 of 9

[7] Comm's

CAN Bus

The CAN Bus circuit diagram shows a microcontroller (FDCAN2) connected to an MCP2551-I-SN CAN transceiver (U12). The transceiver's VDD is connected to a 5V supply through a 100nF capacitor (C88). Its VSS is connected to ground through a 100nF capacitor (C90). The TXD pin of the transceiver is connected to the CAN_TX pin of the microcontroller, and the RXD pin is connected to the CAN_RX pin. The CANH and CANL pins of the transceiver are connected to the CAN+ and CAN- pins of the microcontroller. The CAN+ and CAN- pins are connected to the CAN bus through a 120R resistor (R55) and a 100R resistor (R56) respectively. The CAN bus is terminated at both ends with 120R resistors (R59, R60). The CAN bus is connected to a microcontroller (FDCAN2) through a 4.7K resistor (R52). The microcontroller's TXD and RXD pins are connected to the CAN_TX and CAN_RX pins of the transceiver. The microcontroller's VDD and VSS pins are connected to a 5V supply and ground respectively. The microcontroller's TXD and RXD pins are connected to the CAN_TX and CAN_RX pins of the transceiver. The microcontroller's TXD and RXD pins are connected to the CAN_TX and CAN_RX pins of the transceiver.

RS485/Modbus RTU

The RS485/Modbus RTU circuit diagram shows a microcontroller (RS485) connected to an ST485EDR RS485 transceiver (U14). The transceiver's VCC is connected to a 3.3V supply through a 100nF capacitor (C114). Its GND is connected to ground through a 100nF capacitor (C115). The RXD pin of the transceiver is connected to the RXD pin of the microcontroller, and the TXD pin is connected to the TXD pin. The RS485+ and RS485- pins of the transceiver are connected to the RS485+ and RS485- pins of the microcontroller. The RS485+ and RS485- pins are connected to the RS485 bus through a 120R resistor (R60) and a 120R resistor (R61) respectively. The RS485 bus is terminated at both ends with 120R resistors (R60, R61). The RS485 bus is connected to a microcontroller (RS485) through a 120R resistor (R60). The microcontroller's RXD and TXD pins are connected to the RXD and TXD pins of the transceiver. The microcontroller's RXD and TXD pins are connected to the RXD and TXD pins of the transceiver.

USB2.0 Type-C

The USB2.0 Type-C circuit diagram shows a microcontroller (U17) connected to a TYPE-C 16PLT-H10.0 connector (U17). The connector's VBUS pin is connected to a 5V_USB supply through a 100nF capacitor (C48). Its GND pin is connected to ground through a 100nF capacitor (C49). The connector's DP1, DP2, DN1, and DN2 pins are connected to the DP1, DP2, DN1, and DN2 pins of the microcontroller. The connector's CC1 and CC2 pins are connected to the CC1 and CC2 pins of the microcontroller. The connector's SBU1 and SBU2 pins are connected to the SBU1 and SBU2 pins of the microcontroller. The microcontroller's VDD and VSS pins are connected to a 5V_USB supply and ground respectively. The microcontroller's TXD and RXD pins are connected to the TXD and RXD pins of the microcontroller. The microcontroller's TXD and RXD pins are connected to the TXD and RXD pins of the microcontroller.

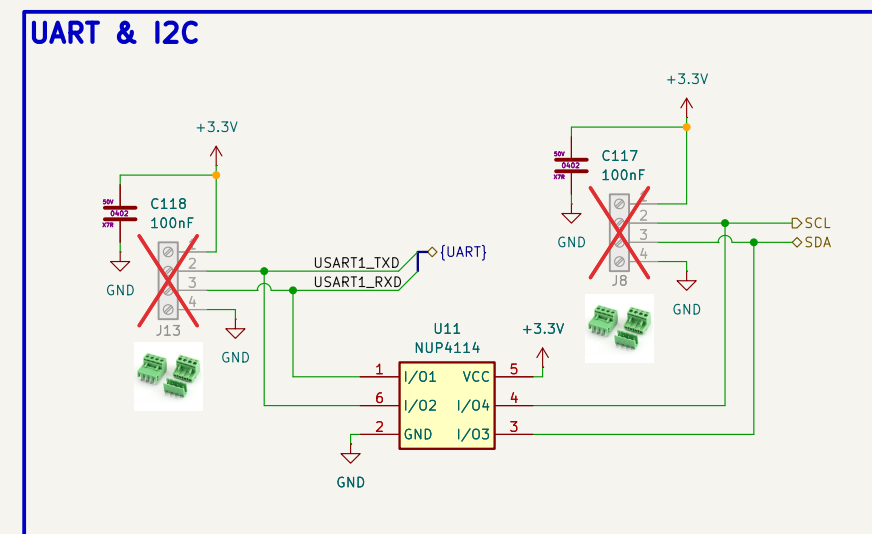
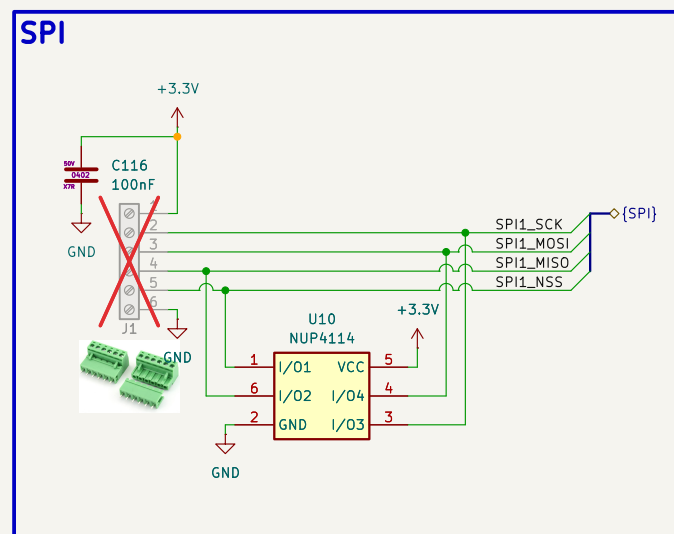
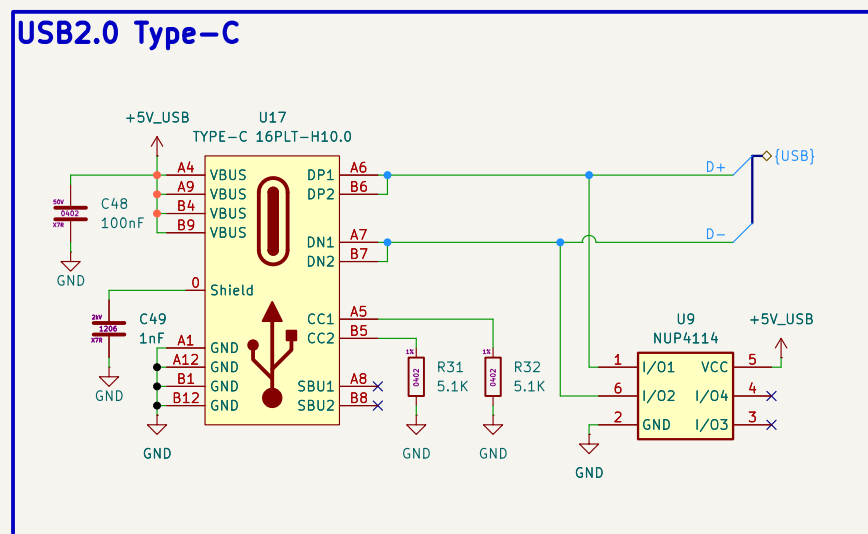
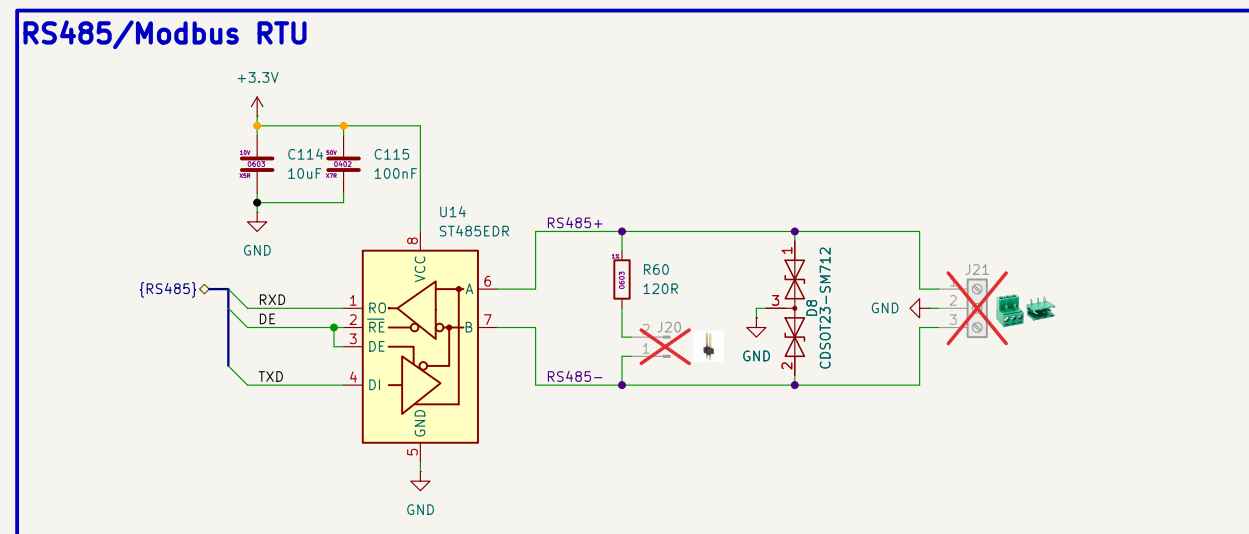
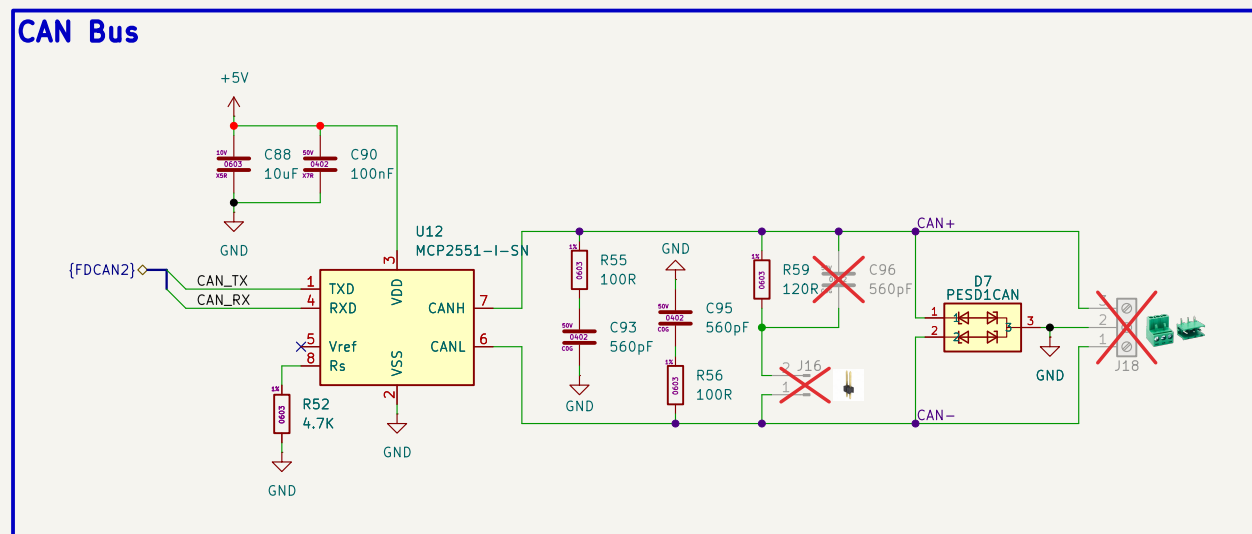
SPI

The SPI circuit diagram shows a microcontroller (U10) connected to an NUP4114 SPI transceiver (U10). The transceiver's VCC is connected to a 3.3V supply through a 100nF capacitor (C116). Its GND is connected to ground through a 100nF capacitor (C117). The SPI1_SCK pin of the transceiver is connected to the SPI1_SCK pin of the microcontroller, and the SPI1_MOSI pin is connected to the SPI1_MOSI pin. The SPI1_MISO pin of the transceiver is connected to the SPI1_MISO pin of the microcontroller, and the SPI1_NSS pin is connected to the SPI1_NSS pin. The SPI1_SCK, SPI1_MOSI, SPI1_MISO, and SPI1_NSS pins are connected to the SPI bus through a 120R resistor (R31) and a 120R resistor (R32) respectively. The SPI bus is terminated at both ends with 120R resistors (R31, R32). The SPI bus is connected to a microcontroller (U10) through a 120R resistor (R31). The microcontroller's SPI1_SCK and SPI1_MOSI pins are connected to the SPI1_SCK and SPI1_MOSI pins of the transceiver. The microcontroller's SPI1_MISO and SPI1_NSS pins are connected to the SPI1_MISO and SPI1_NSS pins of the transceiver.

UART & I2C

The UART & I2C circuit diagram shows a microcontroller (U11) connected to an NUP4114 UART/I2C transceiver (U11). The transceiver's VCC is connected to a 3.3V supply through a 100nF capacitor (C118). Its GND is connected to ground through a 100nF capacitor (C119). The USART1_TXD pin of the transceiver is connected to the USART1_TXD pin of the microcontroller, and the USART1_RXD pin is connected to the USART1_RXD pin. The SCL pin of the transceiver is connected to the SCL pin of the microcontroller, and the SDA pin is connected to the SDA pin. The USART1_TXD, USART1_RXD, SCL, and SDA pins are connected to the UART/I2C bus through a 120R resistor (R31) and a 120R resistor (R32) respectively. The UART/I2C bus is terminated at both ends with 120R resistors (R31, R32). The UART/I2C bus is connected to a microcontroller (U11) through a 120R resistor (R31). The microcontroller's USART1_TXD and USART1_RXD pins are connected to the USART1_TXD and USART1_RXD pins of the transceiver. The microcontroller's SCL and SDA pins are connected to the SCL and SDA pins of the transceiver.

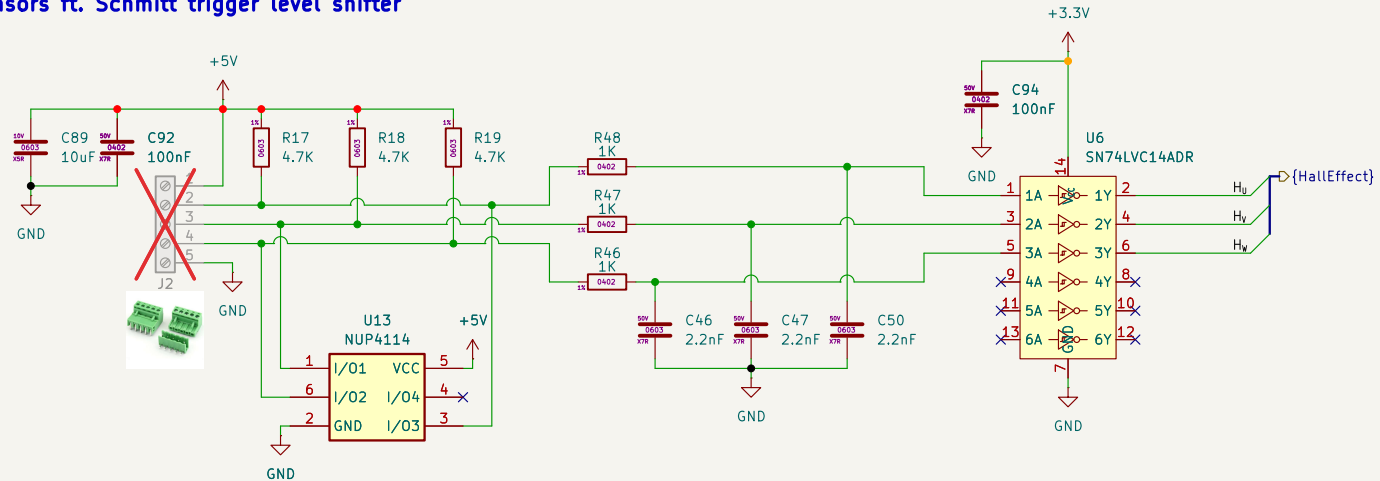
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Sheet Title:		File Name: Comms.kicad_sch	Designer: Bvmy	Date:	Revision: 1.0.0
Sheet Path: /Project Architecture/Comm's/		Reviewer:	Size: A3	Sheet: 7	of 9



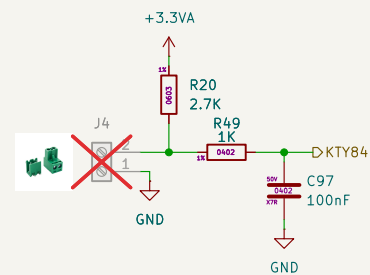
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	Sheet Path: /Project Architecture/Comm's/		Reviewer:	Size: A3	Sheet: 7 of 9	

[8] Motor Output

Hall sensors ft. Schmitt trigger level shifter



Silicon PTC, temp form motor



Comments:

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Board Name:

Vector

Project Name:

Vortex

Sheet Title:

File Name:

MotorOutput.kicad_sch

Designer:

Bvmy

Date:

Revision:

1.0.0

Sheet Path:

/Project Architecture/Motor Output/

Reviewer:

Size:

A4

Sheet:

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A					A
B					B
C					C
D					D
1	2	3	4	5	6

	Comments:	Company: Jovanovski Laboratory - JLab		Variant:	
	Sheet Title:	Board Name: Vector		Project Name: Vortex	
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